

SOLVED PRACTICE WORKBOOK

Space Programme: ISRO, Organisation and Launch Vehicles - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Space Programme: ISRO, Organisation and Launch Vehicles	
REFRESHED TEACHING NAVIGATION — BASIC/CORE FIRST	
01 FOUNDATION — DEPARTMENT AGENCY AND CENTRE ARCHITECTURE Mechanism chain: DoS-ISRO boundary - Centre-function boundary	02 FOUNDATION — IN-SPACE AUTHORISATION AND SUPERVISION Mechanism chain: IN-SPACe boundary Qualified use: Define
03 FOUNDATION — NSIL COMMERCIALISATION BOUNDARY Mechanism chain: NSIL boundary Qualified use: Separate commercial	04 CORE — POLICY TREATY AND DOMESTIC-LAW LAYERS Mechanism chain: Policy-law boundary Qualified use: Distinguish
05 CORE — LAUNCH VEHICLE SPACECRAFT AND PAYLOAD CHAIN A launch vehicle supplies propulsion to place a payload on a	06 CORE — LEO POLAR SUN-SYNCHRONOUS AND GEOSYNCHRONOUS GEOMETRY Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while
07 CORE — TRANSFER ORBIT AND FINAL GEOSTATIONARY SLOT Mechanism chain: PSLV role Qualified use: Show GTO injection before	08 CORE — PSLV MISSION VERSATILITY Mains: Explain PSLV by mission flexibility rather than its label alone.
09 CORE — GSLV CRYOGENIC TRANSFER-ORBIT ROLE LVM3 is India's heavier operational launcher and the base vehicle	10 CORE — LVM3 HEAVY LIFT AND HUMAN-RATING BOUNDARY SSLV is a separate small-satellite launcher designed around smaller
11 CORE — SSLV SMALL-LAUNCH MARKET LOGIC Mains: Explain why SSLV is a distinct vehicle and market architecture.	12 CORE — SLV ASLV OPERATIONAL FLEET AND FUTURE LINEAGE Mains: Use a lineage with retired, operational and future status
13 CORE SYNTHESIS — SOLID LIQUID CRYOGENIC AND SEMI-CRYOGENIC PROPULSION Sovereign access depends on vehicle reliability, manufacturing.	14 CORE SYNTHESIS — LAUNCH CADENCE INFRASTRUCTURE AND MISSION ASSURANCE The Next Generation Launch Vehicle is a Cabinet-approved
15 CORE SYNTHESIS — NGLV PADS TECHNOLOGY TRANSFER AND DATED-STATUS AUDIT Claim: Future ambition is not present capability → NGLV and the Third	

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies DoS-ISRO boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: A. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of DoS-ISRO boundary?

A. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: B. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses DoS-ISRO boundary without changing its institution, unit or status?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: C. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about DoS-ISRO boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: D. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Centre-function boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: A. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Centre-function boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: B. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Centre-function boundary without changing its institution, unit or status?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: C. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not

transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Centre-function boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.

Answer: D. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies IN-SPACE boundary?

A. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: A. Explanation: IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of IN-SPACE boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: B. Explanation: IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses IN-SPACE boundary without changing its institution, unit or status?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: C. Explanation: IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about IN-SPACe boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: D. Explanation: IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies NSIL boundary?

A. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: A. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of NSIL boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: B. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses NSIL boundary without changing its institution, unit or status?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.

Answer: C. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about NSIL boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: D. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Policy-law boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: A. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Policy-law boundary?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: B. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Policy-law boundary without changing its institution, unit or status?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.

Answer: C. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Policy-law boundary?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: D. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Launcher-spacecraft boundary?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: A. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Launcher-spacecraft boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: B. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Launcher-spacecraft boundary without changing its institution, unit or status?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: C. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Launcher-spacecraft boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: D. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Orbit-family boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: A. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Orbit-family boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: B. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms

for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Orbit-family boundary without changing its institution, unit or status?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: C. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Orbit-family boundary?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.

Answer: D. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies GTO-GEO boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: A. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of GTO-GEO boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: B. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the

spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses GTO-GEO boundary without changing its institution, unit or status?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: C. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about GTO-GEO boundary?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: D. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies PSLV role?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: A. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of PSLV role?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: B. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses PSLV role without changing its institution, unit or status?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: C. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about PSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.

Answer: D. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies GSLV role?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of GSLV role?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: B. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses GSLV role without changing its institution, unit or status?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: C. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about GSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: D. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies LVM3 role?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of LVM3 role?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: B. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses LVM3 role without changing its institution, unit or status?

A. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: C. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about LVM3 role?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: D. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies SSLV role?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of SSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: B. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses SSLV role without changing its institution, unit or status?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: C. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about SSLV role?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: D. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Lineage-status boundary?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: A. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Lineage-status boundary?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: B. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Lineage-status boundary without changing its institution, unit or status?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: C. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Lineage-status boundary?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: D. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Propulsion-class boundary?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: A. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Propulsion-class boundary?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: B. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Propulsion-class boundary without changing its institution, unit or status?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: C. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Propulsion-class boundary?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: D. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Human-rating boundary?

A. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: A. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Human-rating boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Human-rating boundary without changing its institution, unit or status?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: C. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Human-rating boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: D. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Launch-system boundary?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: A. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Launch-system boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Launch-system boundary without changing its institution, unit or status?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: C. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Launch-system boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: D. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies NGLV status boundary?

- A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: A. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of NGLV status boundary?

- A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses NGLV status boundary without changing its institution, unit or status?

- A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: C. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about NGLV status boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: D. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Infrastructure-status boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: A. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Infrastructure-status boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: B. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Infrastructure-status boundary without changing its institution, unit or status?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: C. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Infrastructure-status boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: D. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Technology-transfer boundary?

A. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: A. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Technology-transfer boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.

Answer: B. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Technology-transfer boundary without changing its institution, unit or status?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: C. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Technology-transfer boundary?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: D. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile launch-data boundary?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: A. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile launch-data boundary?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: B. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile launch-data boundary without changing its institution, unit or status?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: C. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile launch-data boundary?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: D. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2018 PSLV-GSLV objective concept and a provisional 2026 IN-SPACe/private-participation concept here. No unverified objective key or direct Mains demand is invented.

PYQ DEMAND CARD 1 - 2018 Prelims GS-I

Demand: Compare PSLV and GSLV as Indian satellite launch vehicles.

Status: Verified routed objective concept; the official key was unavailable locally, so no answer letter is asserted.

Model solution: Launcher-spacecraft boundary: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **GTO-GEO boundary:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **PSLV role:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **GSLV role:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2018 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Launcher-spacecraft boundary: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **GTO-GEO boundary:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **PSLV role:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **GSLV role:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Compare PSLV and GSLV as Indian satellite launch vehicles. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed objective concept; the official key was unavailable locally, so no answer letter is asserted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Launcher-spacecraft boundary: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **GTO-GEO boundary:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **PSLV role:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **GSLV role:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2018 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2026 Prelims GS-I

Demand: Private-sector participation, IN-SPACe mandate and Indian space-startup achievements.

Status: Verified provisional routed concept; the local key is provisional and no answer letter is recorded or inferred.

Model solution: DoS-ISRO boundary: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **IN-SPACe boundary:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **NSIL boundary:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Policy-law**

boundary: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Technology-transfer boundary:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 2 - 2026 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: DoS-ISRO boundary: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **IN-SPACe boundary:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **NSIL boundary:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Policy-law boundary:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Technology-transfer boundary:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Private-sector participation, IN-SPACe mandate and Indian space-startup achievements. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified provisional routed concept; the local key is provisional and no answer letter is recorded or inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: DoS-ISRO boundary: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **IN-SPACe boundary:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **NSIL boundary:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Policy-law boundary:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Technology-transfer boundary:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish the roles of DoS, ISRO, IN-SPACe and NSIL. Answer in about 150 words.

Model thesis: **Claim:** DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.
- VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.
- IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.
- NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.
- Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Qualified conclusion: **Claim:** DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is

the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish the roles of DoS, ISRO, IN-SPACe and NSIL. Answer in about 150 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer

the mandate of the whole agency to it. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

5. **Claim and named evidence:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish the roles of DoS, ISRO, IN-SPACe and NSIL. Answer in about 150 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate PSLV, GSLV, LVM3 and SSLV without reducing the comparison to payload size. Answer in about 150 words.

Model thesis: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role.

Named evidence/example: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.
- Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.
- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Qualified conclusion: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system

boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Differentiate PSLV, GSLV, LVM3 and SSLV without reducing the comparison to payload size....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or

experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy

implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Differentiate PSLV, GSLV, LVM3 and SSLV without reducing the comparison to payload size....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain how propulsion and orbit requirements shape launcher choice. Answer in about 250 words.

Model thesis: **Claim:** Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for

Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.
- LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.
- Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.
- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Qualified conclusion: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary.

Named evidence/example: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named**

evidence/example: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named**

evidence/example: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy

implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how propulsion and orbit requirements shape launcher choice. Answer in about 250...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary.

Named evidence/example: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named**

evidence/example: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named**

evidence/example: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary.

Named evidence/example: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or

experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how propulsion and orbit requirements shape launcher choice. Answer in about 250...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess Indian space reform as institutional differentiation rather than privatisation. Answer in about 250 words.

Model thesis: **Claim:** DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.
- IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.
- NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.
- Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Qualified conclusion: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **assess** requires a direct position on "Assess Indian space reform as institutional differentiation rather than privatisation. Answer...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector

commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or

experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Assess Indian space reform as institutional differentiation rather than privatisation. Answer...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's access-to-space ecosystem through vehicles, infrastructure, industry and mission assurance. Answer in about 300 words.

Model thesis: **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the

scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPaCe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.
- Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.
- The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPaCe, NSIL and the State's safety responsibilities disappear.
- Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Qualified conclusion: **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before

drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **evaluate** requires a direct position on "Evaluate India's access-to-space ecosystem through vehicles, infrastructure, industry and...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The

conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPaCe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Evaluate India's access-to-space ecosystem through vehicles, infrastructure, industry and...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a status-disciplined roadmap from the current fleet to future launch capacity. Answer in about 300 words.

Model thesis: Claim: Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Human-rating boundary. **Named evidence/example:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NGLV status boundary. **Named evidence/example:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.
- Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.
- Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

- Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.
- The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.
- The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.
- Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Qualified conclusion: Claim: Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Human-rating boundary. **Named evidence/example:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NGLV status boundary. **Named evidence/example:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary.

Named evidence/example: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Design a status-disciplined roadmap from the current fleet to future launch capacity. Answer...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Human-rating boundary. **Named evidence/example:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:**

Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NGLV status boundary. **Named evidence/example:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:**

Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary.

Named evidence/example: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a

flying cryogenic stage. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Design a status-disciplined roadmap from the current fleet to future launch capacity. Answer...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.